Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

A stationary pulley carries a rope one end of which supports a ladder with a man and the other a counter weight of mass M. A man of mass m climbs up a distance l w.r.t the ladder and then stops. The displacement of the centre of mass of this system is

(A) $\frac{ml}{M+m}$

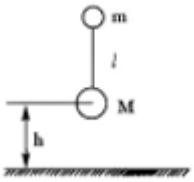
(B) $\frac{ml}{2M}$

(C) $\frac{ml}{M+2m}$

(D) $\frac{ml}{2M+m}$

Q.2

A small ball of mass m is connected by an inextensible massless string of length l with an another ball of mass $M = 4\ \mathrm{m}$. They are released with zero tension in the string from a height h as shown in the figure. Find the time when the string becomes taut for the first time after the mass M collides with the ground. Take all the collisions to be elastic.



(A) $\frac{l}{\sqrt{3gh}}$

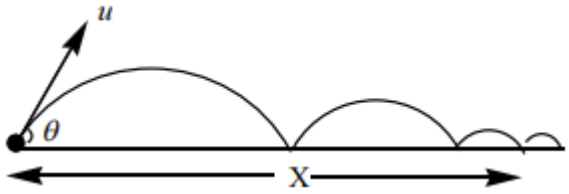
(B) $\frac{l}{\sqrt{2gh}}$

(C) $\frac{\sqrt{3l}}{2gh}$

(D) $\frac{l}{\sqrt{gh}}$

Q.3

A ball is projected with initial velocity u at an angle θ to the horizontal. Then horizontal displacement covered by ball as it collides third time to the ground would be, if coefficient of restitution is e:



(A) $(1+e)\frac{u^2 \sin 2\theta}{g}$

(B) $e\frac{u^2 \sin 2\theta}{g}$

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(C) $(1-e)\frac{u^2 \sin 2\theta}{g}$

(D) $\left(1+e+e^2\right)\frac{u^2 \sin 2\theta}{g}$

Q.4

Two putty balls of equal mass moving with equal velocity in mutually perpendicular directions, stick together after collision. If the balls were initially moving with a velocity of $45\sqrt{2}\ \mathrm{ms}^{-1}$ each, the velocity of their combined mass after collision is

(A) $45\sqrt{2}\ \mathrm{ms}^{-1}$

(B) $45\ \mathrm{ms}^{-1}$

(C) $90\ \mathrm{ms}^{-1}$

(D) $22.5\sqrt{2}\ \mathrm{ms}^{-1}$

Q.5

A tennis ball is released from height h above ground level. If the ball makes inelastic collision with the ground, to what height will it rise after third collision

(A) he^6

(B) $\mathrm{e}^2\ \mathrm{h}$

(C) $\mathrm{e}^3\ \mathrm{h}$

(D) None of these

Q.6

A ball is projected vertically down with an initial velocity from a height of $20\ \mathrm{m}$ onto a horizontal floor. During the impact it loses 50% of its energy and rebounds to the same height. The initial velocity of its projection is

(A) $20\ \mathrm{m/s}$

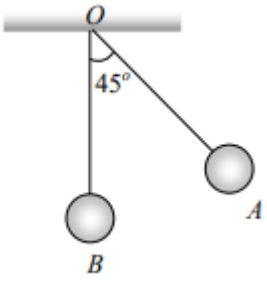
(B) $15\ \mathrm{m/s}$

(C) $10\ \mathrm{m/s}$

(D) $5\ \mathrm{m/s}$

Q.7

The bob A of a simple pendulum is released when the string makes an angle of 45° with the vertical. It hits another bob B of the same material and same mass kept at rest on the table. If the collision is elastic -



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(A) Both A and B rise to the same height

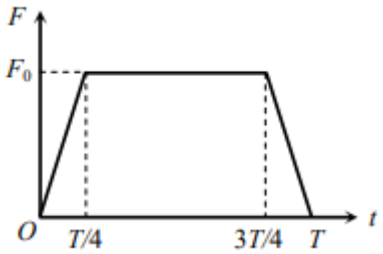
(B) Both A and B come to rest at B

(C) Both A and B move with the same velocity of A

(D) A comes to rest and B moves with the velocity of A

Q.8

A particle of mass m moving with a velocity u makes an elastic one dimensional collision with a stationary particle of mass m establishing a contact with it for extremely small time T. Their force of contact increases from zero to F_0 linearly in time $\mathrm{T}/4$, remains constant for a further time $\mathrm{T}/2$ and decreases linearly from F_0 to zero in further time $\mathrm{T}/4$ as shown. The magnitude possessed by F_0 is -



(A) mu/T

(B) $2\mathrm{mu}/\mathrm{T}$

(C) $4\mathrm{mu}/3\ \mathrm{T}$

(D) $3\mathrm{mu}/4\ \mathrm{T}$

Q.9

A set of n identical cubical blocks lies at rest parallel to each other along a line on a smooth horizontal surface. The separation between the near surfaces of any two adjacent blocks is L. The block at one end is given a speed v towards the next one at time $t = 0$. All collisions are completely inelastic, then

(A) The last block starts moving at $t = \frac{(n-1)L}{v}$

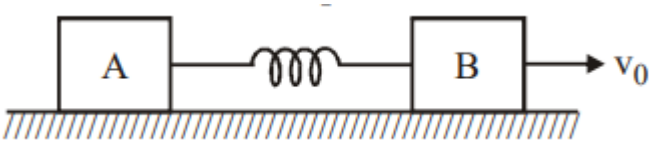
(B) The last block starts moving at $\mathrm{t} = \frac{\mathrm{n}(\mathrm{n}-1)\mathrm{L}}{2\mathrm{v}}$

(C) The centre of mass of the system will have a final speed v

(D) The centre of mass of the system will have a final speed v/n

Q.10

Two blocks A and B of the same mass are connected to a light spring and placed on a smooth horizontal surface. B is given velocity v_0 (as shown in figure) when the spring is in natural length. In the subsequent motion. Choose the correct options -



(A) the maximum velocity of B will be v_0

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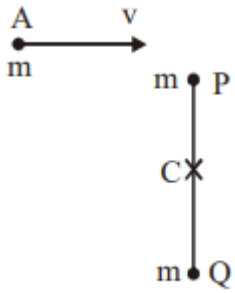
(B) as seen from ground, A can move towards right only

(C) the spring will be at natural length again when B is at rest

(D) None of these

Q.11

Two particles P and Q of mass m each are at rest and connected by a rigid rod of length L and negligible mass. Another particle A of mass m moving at a speed v normal to PQ strikes P and sticks to it. The centre of mass of $(\mathrm{P}+\mathrm{Q}+\mathrm{A})$ system is C.



(A) The velocity of centre of mass C before impact is $\mathrm{v}/3$.

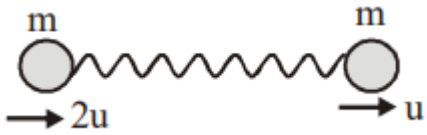
(B) The velocity of C after impact is $\mathrm{v}/3$.

(C) The angular velocity of the system about C is $\mathrm{v}/2\ \mathrm{L}$

(D) The moment of inertia of the P, Q and A system about C is $\frac{2}{3}mL^2$

Q.12

Passage:
Two identical balls, each of mass m, are tied with a string and kept on a frictionless surface. Initially the string is slack. They are given velocities $2\mathrm{u}$ and u in the same direction. Collision between the balls is perfectly elastic.



Question: After the first collision, what is the total loss in kinetic energy of the balls ?

(A) $2\mathrm{mu}^2$

(B) mu^2

(C) $3\mathrm{mu}^2$

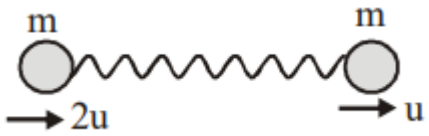
(D) zero

Q.13

Passage:
Two identical balls, each of mass m, are tied with a string and kept on a frictionless surface. Initially the string is

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slack. They are given velocities $2u$ and u in the same direction. Collision between the balls is perfectly elastic.



Question: What is the final loss in kinetic energy of the balls -

(A) $mu^2/4$

(B) $mu^2/2$

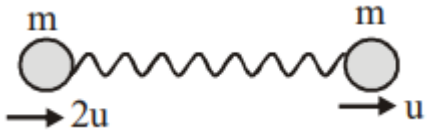
(C) $3mu^2/4$

(D) None of these

Q.14

Passage:

Two identical balls, each of mass m, are tied with a string and kept on a frictionless surface. Initially the string is slack. They are given velocities $2u$ and u in the same direction. Collision between the balls is perfectly elastic.



Question: What is the impulse generated in the string during the second collision -

(A) $mu/2$

(B) mu/L

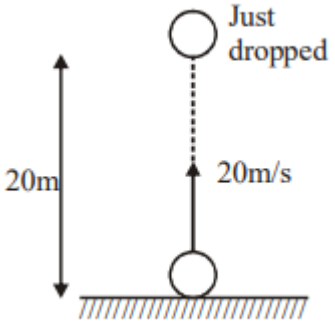
(C) $2mu/3$

(D) None of these

Q.15

Passage:

Two identical masses are as shown in figure. One is thrown upwards with velocity $20\ m/s$ and another is just dropped simultaneously.



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Question: The masses collide in air and stick together. After how much time the combined mass will fall to the ground (calculate the time from the starting when the motion was started) -

(A) $(1+\sqrt{2})$s

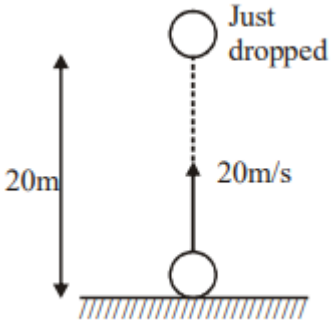
(B) $2\sqrt{2}$ s

(C) $(2+\sqrt{2})$s

(D) None of these

Q.16

Passage:
Two identical masses are as shown in figure. One is thrown upwards with velocity $20\ \mathrm{m/s}$ and another is just dropped simultaneously.



Question: In the above problem, to what maximum height (from ground) will the combined mass rise -

(A) $25\ \mathrm{m}$

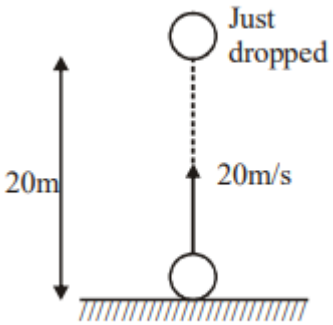
(B) $18\ \mathrm{m}$

(C) $15\ \mathrm{m}$

(D) $20\ \mathrm{m}$

Q.17

Passage:
Two identical masses are as shown in figure. One is thrown upwards with velocity $20\ \mathrm{m/s}$ and another is just dropped simultaneously.



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Question: If collision between them is elastic, find the time interval between their striking with ground -

(A) zero

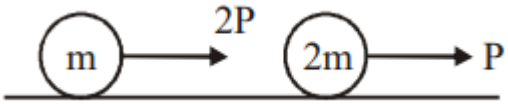
(B) $2\ \mathrm{s}$

(C) $1\ \mathrm{s}$

(D) $3\ \mathrm{s}$

Q.18

Two balls of mass m and $2\ \mathrm{m}$ each have momentum $2\mathrm{P}$ and P in the direction shown in figure. During collision they exert an impulse of magnitude P on each other. Match the following:



Column I	**Column II**
(A) After collision momentum of m	(p) 2P
(B) After collision momentum of 2m	(q) P
(C) Coefficient of restitution between them	(r) 1
	(s) None

(A) A-(q) B-(s) C-(p)

(B) A-(p) B-(q) C-(r)

(C) A-(q) B-(p) C-(s)

(D) A-(p) B-(p) C-(s)

Q.19

Neutron Decay. A neutron at rest decays (breaks up) to a proton and an electron. Energy is released in the decay and appears as kinetic energy of the proton and electron. The mass of a proton is 1836 times the mass of an electron. What fraction of the total energy released goes into the kinetic energy of the proton? Mark answer in percentage.

Q.20

Direct central impact occurs between a $300\ \mathrm{N}$ body (a) moving to the right with a velocity of $6\ \mathrm{m/s}$ and $150\ \mathrm{N}$ body (b) moving to the left with a velocity of $10\ \mathrm{m/s}$. Given the coefficient of restitution is 0.8. Then, after impact, we have the velocities as $|\mathrm{V}'_{\mathrm{a}}| = \mathrm{xm/s}$ and a will be moving in left direction, $|\mathrm{V}'_{\mathrm{b}}| = \mathrm{ym/s}$ and b will be moving in right direction then find the value of $(\mathrm{x}+\mathrm{y})10 - 120$.

Answers & Solutions

Q.1 Answer:

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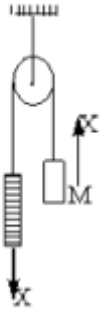
$\frac{ml}{2M}$

Solution:

Mass of $\mathrm{man} = \mathrm{m}$

Mass of ladder $= \mathrm{M} - \mathrm{m}$

$$\Delta x_{CM} = \frac{Mx - (M-m)x + m(l-x)}{2M}$$
$$= \frac{Mx + ml - mx}{2M} = \frac{ml}{2M}$$



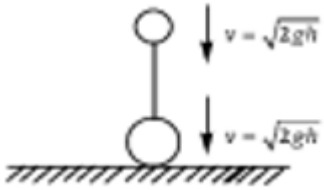
Q.2 Answer:

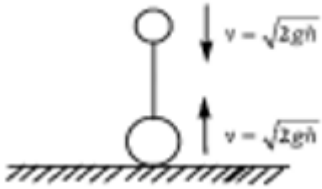
$\frac{l}{\sqrt{2gh}}$

Solution:

Just after collision velocities of both the balls will be $\sqrt{2g}$ **in opposite directions Relative acceleration between the two balls is zero and relative velocity of approach is** $2\sqrt{2ght}$**. Hence they will collide after a time**

$t = \frac{1}{2\sqrt{2gh}}$





Hence relative velocity of separation will be $2\sqrt{2gh}$ **(collision is elastic)**

Hence the string becomes tight after the same time, $t - \frac{1}{2\sqrt{2gh}}$

Hence the total time will be $\frac{l}{\sqrt{2gh}}$

Q.3 Answer:

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$\left(1+e+e^2\right) \frac{u^2 \sin 2\theta}{g}$

Solution:

After each collision vertical velocity become e times whereas horizontal velocity remain same. So after each collision time of flight become e times of previous one.

So that horizontal displacement $= R + eR + e^2R = \left(1+e+e^2\right) \frac{u^2 \sin 2\theta}{g}$

Q.4 Answer:

45 ms^{-1}

Solution:

Initial momentum $\overrightarrow{\mathrm{P}} = \mathrm{m} \times 45\sqrt{2}\hat{i} + \mathrm{m} \times 45\sqrt{2}\hat{j} \Rightarrow |\overrightarrow{\mathrm{P}}| = \mathrm{m} \times 90$

Final momentum $2\ \mathrm{m} \times \mathrm{V}$

By conservation of momentum $2\ \mathrm{m} \times \mathrm{V} = \mathrm{m} \times 90$

$\therefore \quad V = 45\ \mathrm{m/s}$

Q.5 Answer:

he^6

Solution:

Given that,

Height $= \mathrm{h}$**, Final velocity** $= v$**, Initial velocity** $= u$

Coefficient restitution $= \mathrm{e}$

We know that,

Restitution = the square root of the ratio of the kinetic energy before and after collision

$$\mathrm{e} = \frac{\mathrm{v}}{\mathrm{u}} = \sqrt{\frac{\mathrm{K}_1}{\mathrm{K}_0}}$$

$$\mathrm{K}_1 = \mathrm{e}^2\ \mathrm{K}_0$$

$$\mathrm{K}_2 = \mathrm{e}^2\ \mathrm{K}_1$$

$$\mathrm{K}_2 = \mathrm{e}^2\ \mathrm{K}_0$$

Similarly

$$\mathrm{K}_3 = \mathrm{e}^4\ \mathrm{K}_2$$

$$\mathrm{K}_3 = \mathrm{e}^6\ \mathrm{K}_0$$

We know that, The kinetic energy is

$$\mathrm{mgh} = \mathrm{K}$$

$$\mathrm{h} = \mathrm{K}/\mathrm{mg}$$

The ball is initially dropped from the height hand acquires a kinetic energy K_0 when it hits the ground, before the first collision.

$$\mathrm{mgh} = \mathrm{K}_0$$

Now, If h_1, h_2 and h_3 are the heights after the first, second and third collisions

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$h_1 = K_1/mg = e^2\ K_0/mg = e^2\ h$

$h_2 = e^4\ h,\ h_3 = e^6\ h$

Hence, the height raised by the ball after third collision is $e^6\ h$

Q.6 Answer:

$20\ m/s$

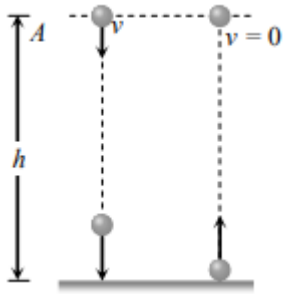
Solution:

Let ball is projected vertically downward with velocity v **from height** h **Total energy at point** $A = \frac{1}{2}mv^2 + mgh$

During collision loss of energy is 50% **and the ball rises up to same height. It means it possess only potential energy at same level.**

$50\%\ \left(\frac{1}{2}mv^2 + mgh\right) = mgh;\ \frac{1}{2}\left(\frac{1}{2}mv^2 + mgh\right) = mgh;\ v = \sqrt{2gh} = \sqrt{2 \times 10 \times 20}$

$\therefore \quad V = 20\ m/s$



Q.7 Answer:

A comes to rest and B moves with the velocity of A

Solution:

Due to the same mass of A **and** B **as well as due to elastic collision velocities of spheres get interchanged after the collision.**

Q.8 Answer:

$4mu/3\ T$

Solution:

Change in momentum = Impulse = Area under force-time graph

$\therefore \quad mv =$ **Area of trapezium**

$\Rightarrow \quad mv = \frac{1}{2}\left(T + \frac{T}{2}\right)F_0 \Rightarrow mv = \frac{3\,T}{4}F_0 \Rightarrow F_0 = \frac{4mu}{3\,T}$

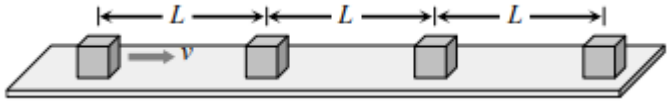
Q.9 Answer:

The last block starts moving at $t = \frac{n(n-1)L}{2v}$

Solution:

Since collision is perfectly inelastic so all the blocks will stick together one by one and move in a form of combined mass.

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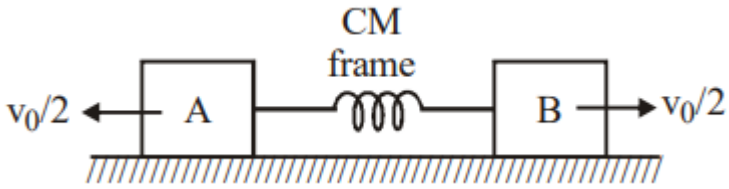
Time required to cover a distance ' L ' by first block $= \frac{L}{v}$
Now first and second block will stick together and move with v/2 velocity (by applying conservation of momentum) and combined system will take time $\frac{\mathrm{L}}{\mathrm{V}/2} = \frac{2\,\mathrm{L}}{\mathrm{V}}$ to reach up to block third.
Now these three blocks will move with velocity v/3 and combined system will take time $\frac{\mathrm{L}}{\mathrm{v}/3} = \frac{3\,\mathrm{L}}{\mathrm{v}}$ to reach upto the block fourth.
So, total time $= \frac{\mathrm{L}}{\mathrm{v}} + \frac{2\,\mathrm{L}}{\mathrm{v}} + \frac{3\,\mathrm{L}}{\mathrm{v}} + \ldots.. + \frac{(\mathrm{n}-1)\mathrm{L}}{\mathrm{v}} = \frac{\mathrm{n}(\mathrm{n}-1)\mathrm{L}}{2\mathrm{v}}$ and velocity of combined system having n blocks as $\frac{\mathrm{v}}{\mathrm{n}}$.

Q.10 Answer:

the maximum velocity of B will be v_0

Solution:

$\mathrm{v}_{\mathrm{CM}} = \frac{\mathrm{v}_0}{2}$



Both blocks can have maximum velocity $\mathrm{v}_0/2$ towards right in CM frame.
Thus, $(\mathrm{v}_\mathrm{B})_{\max} = \mathrm{v}_0$ in ground frame.
$(\mathrm{v}_\mathrm{A})_{\min} = 0$ in ground frame when A has velocity $\mathrm{v}_0/2$ towards left in CM frame.
Also, In CM frame, $\mathrm{v}_\mathrm{B} = \mathrm{v}_0/2$ left in the case when spring is at natural length.
Thus, $\mathrm{v}_\mathrm{B} = 0$ in ground frame at that instant.

Q.11 Answer:

The velocity of centre of mass C before impact is $\mathrm{v}/3$.

Solution:

$\mathrm{v}_{\mathrm{cm}} = \frac{\mathrm{m}_1\mathrm{u}_1+\mathrm{m}_2\mathrm{u}_2+\mathrm{m}_3\mathrm{u}_3}{\mathrm{m}_1+\mathrm{m}_2+\mathrm{m}_3} = \frac{\mathrm{mv}+0+0}{\mathrm{m}+\mathrm{m}+\mathrm{m}} = \frac{\mathrm{v}}{3}$
Since there is no external torque on the system $\overrightarrow{\mathrm{v}}_{\mathrm{cm}} = \overrightarrow{\mathrm{u}}_{\mathrm{cm}} = \frac{\mathrm{v}}{3}$
After impact, centre of mass of the system is at C
$\mathrm{PC}(2\ \mathrm{m}) = \mathrm{m}(\mathrm{L} - \mathrm{PC}) \Rightarrow \mathrm{PC} = \frac{\mathrm{L}}{3}$
M.I. of the system $(P + Q + A)$ about $C = 2m\left(\frac{L}{3}\right)^2 + m\left(\frac{2L}{3}\right)^2 = \frac{2mL^2}{9} + \frac{4mL^2}{9} = \frac{2}{3}mL^2$
Applying angular momentum conservation about C
$\mathrm{mv}\frac{\mathrm{L}}{3} = \frac{2}{3}\,\mathrm{mL}^2\omega; \omega = \frac{\mathrm{v}}{2\,\mathrm{L}}$

Q.12 Answer:

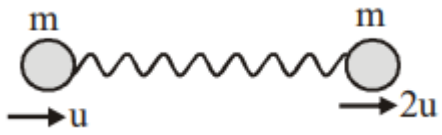
zero

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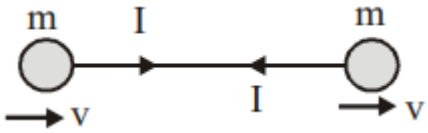
Solution:

(i) Since the collision is elastic, there is no loss in K.E.

(ii) After first collision, the velocities are as shown in figure.



Now after some time, the string will become tight. Let at this point the common velocity of the ball is v.



$2mv = mu + 2mu \Rightarrow v = 3u/2$

$KE_i = \frac{1}{2}mu^2 + \frac{1}{2}\,m(2u)^2 = \frac{5mu^2}{2}; KE_f = \frac{1}{2}mv^2 = \frac{1}{2}2\,m\left(\frac{3u}{2}\right)^2 = \frac{9mu^2}{4}$

$KE_{Loss} = KE_i - KE_f = \frac{5mu^2}{2} - \frac{9mu^2}{4} = \frac{mu^2}{4}$

(iii) For left ball : $I = mv - mu = m\frac{3u}{2} - mu = \frac{mu}{2}$

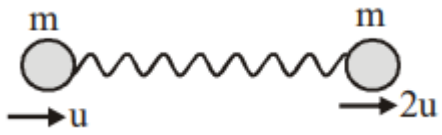
Q.13 Answer:

$mu^2/4$

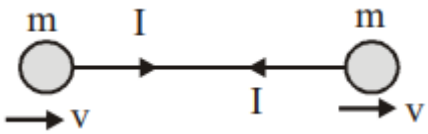
Solution:

(i) Since the collision is elastic, there is no loss in K.E.

(ii) After first collision, the velocities are as shown in figure.



Now after some time, the string will become tight. Let at this point the common velocity of the ball is V.



$2mv = mu + 2mu \Rightarrow v = 3u/2$

$KE_i = \frac{1}{2}mu^2 + \frac{1}{2}\,m(2u)^2 = \frac{5mu^2}{2}; KE_f = \frac{1}{2}mv^2 = \frac{1}{2}2\,m\left(\frac{3u}{2}\right)^2 = \frac{9mu^2}{4}$

$KE_{Loss} = KE_i - KE_f = \frac{5mu^2}{2} - \frac{9mu^2}{4} = \frac{mu^2}{4}$

(iii) For left ball : $I = mv - mu = m\frac{3u}{2} - mu = \frac{mu}{2}$

Q.14 Answer:

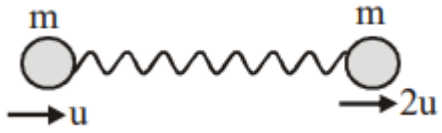
$mu/2$

Solution:

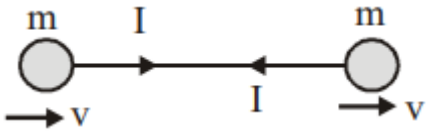
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$KE_{Loss} = KE_i - KE_f = \frac{5mu^2}{2} - \frac{9mu^2}{4} = \frac{mu^2}{4}$

(iii) For left ball : $I = mv - mu = m\frac{3u}{2} - mu = \frac{mu}{2}$

Q.15 Answer:

None of these

Solution:

$v_{CM} = \frac{m\times 20 + m\times 0}{2\,m} = 10\ m/s; t = \frac{2u}{g} = 2sec$

Q.16 Answer:

$15\ m$

Solution:

$H = \frac{u^2}{2\,g} = \frac{10\times 10}{2\times 10} = 5\ m$

Initial height of C.M. $= \frac{m\times 0 + m\times 20}{2\,m} = 10\ m$

Total height $= 10 + 5 = 15\ m$

Q.17 Answer:

$2\ s$

Solution:

In elastic collision between two identical masses, the masses just exchange their velocities. So time would remain same as would had been it they had not collides.

$t_1 = \sqrt{\frac{2\,h}{g}} = \sqrt{\frac{2\times 20}{10}} = 2\ s; t_2 = \frac{2u}{g} = \frac{2\times 20}{10} = 4\ s$

$\Delta t = t_2 - t_1 = 2\ s$

Q.18 Answer:

A-(q) B-(p) C-(s)

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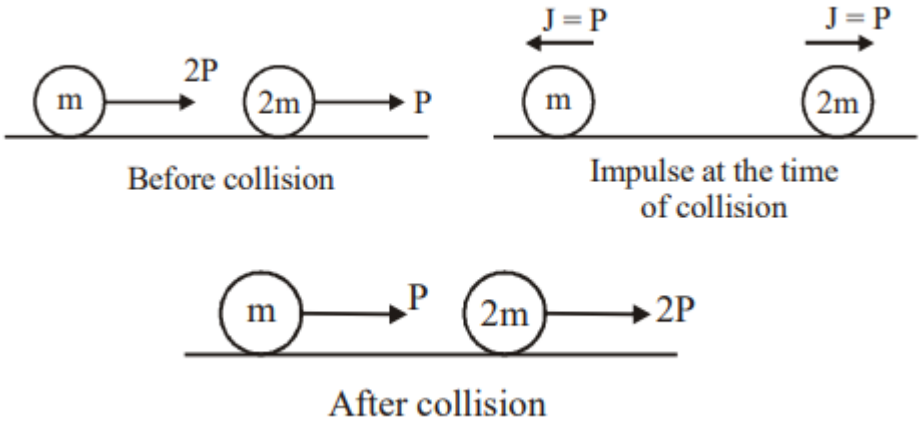
Solution:

$(A) - q, (B) - p, (C) - s.$

$$e = \frac{\text{Relative velocity of separation}}{\text{Relative velocity of approach}}$$

$$\frac{v_{2m} - v_m}{v_m - v_{2m}} = \frac{\left(\frac{2P}{2m}\right) - \left(\frac{P}{m}\right)}{\frac{2P}{m} - \frac{P}{2m}} = 0$$

i.e., they will stick together.



Q.19 Answer:

—

Solution:

KEY & IDENTIFY: Apply conservation of momentum to the system of the neutron and its decay products.

SET UP: Let the proton be moving in the $+x$-direction with speed v_p after the decay. The initial momentum of the neutron is zero, so to conserve momentum the electron must be moving in the $-x$-direction after the decay. Let the speed of the electron be v_e.

EXECUTE: $P_{1x} = P_{2x}$ gives $0 = m_p v_p - m_e v_e$ and $v_e = \left(\frac{m_p}{m_e}\right) v_p$. The total kinetic energy after the decay is $K_{tot} = \frac{1}{2} m_e v_e^2 + \frac{1}{2} m_p v_p^2 = \frac{1}{2} m_e \left(\frac{m_p}{m_e}\right)^2 v_p^2 + \frac{1}{2} m_p v_p^2 = \frac{1}{2} m_p v_p^2 \left(1 + \frac{m_p}{m_e}\right)$

Thus, $\frac{K_p}{K_{tot}} = \frac{1}{1 + m_p/m_e} = \frac{1}{1+1836} = 5.44 \times 10^{-4} = 0.0544\%$.

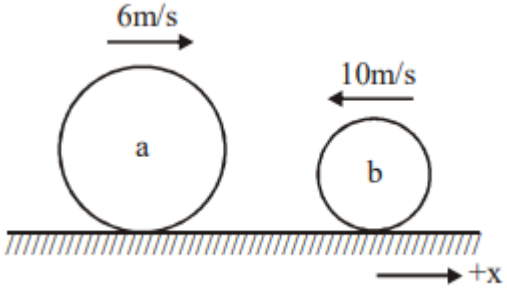
EVALUATE: Most of the released energy goes to the electron, since it is much lighter than the proton.

Q.20 Answer:

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Solution:

$$m_a = \frac{300}{g}\ \text{kg},\ m_b = \frac{150}{g}\ \text{kg}$$



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Before impact:

$V_a = 6\ m/s$ **(defining rightwards as + ve direction)**

$V_b = -10\ m/s$

Apply, $m_a V_a + m_b V_b = m_a V'_a + m_b V'_b$

$\Rightarrow \quad 2\ V'_a + V'_b = 2 ... (1)$

$e = -\left(\frac{V'_b - V'_a}{V_b - V_a}\right) \Rightarrow 0.8 = -\left(\frac{V'_b - V'_a}{-10-6}\right) \Rightarrow V'_a - V'_b = 12.8 ... (2)$

Solving eq. (1) and (2)

$\Rightarrow V'_a = -3.6\ m/s, V_b' = 9.2\ m/s$

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